

Requirements

Python

- Python **3.10 or later** (developed and tested on Python 3.13.7).

Operating systems

Tested on macOS (Darwin 25.4.0). The tool is pure Python and should run on Linux and Windows without modification.

Python package dependencies

Package	Minimum version	Purpose
numpy	1.24	Numerical arrays
pandas	2.0	CSV loading
scikit-learn	1.3	LR / RF / GBM models, train/test split
scipy	1.10	Wilcoxon rank-sum statistical test
matplotlib	3.7	Box plots and convergence curves
joblib	1.3	Persisting trained sklearn models
tensorflow	2.15	Loading the pre-trained DNN .h5 models
keras	3.0	Keras API surface (bundled with TF 2.x)
pytest	8.0	Unit-test runner for <code>tests/</code>

Install everything in one step:

```
pip install -r requirements.txt
```

Hardware

No GPU required. The full 30-run experiment matrix completes in under two hours on a 2024 MacBook Pro (M-series CPU). DNN inference dominates runtime; sklearn families finish in under a minute each.

Data and models

The `DNN/` and `dataset/` directories contain the pre-trained Keras .h5 models and processed CSV datasets sourced from ideas-labo/ISE/lab4. No additional downloads are required.